Cao Thang Nguyen

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EDUCATION

Ph.D. Mechanical Engineering,
Ulsan National Institute of Science & Technology (UNIST), Korea, 2022

M.S. Mechanical Engineering,
Univertisty of Ulsan, Korea, 2016

B.S. Mechanical Engineering,
Univertisty of Technical & Education, Vietnam, 2010

RESEARCH AREAS

Computational mechanics and materials science.

Modeling of material, structural elastic instabilities and phase transformation.

Molecular Dynamics Simulation, Enhanced Sampling Simulations.

TECHNICAL SKILLS

Experienced in atomistic simulations, molecular **dynamics/statics** simulations, transition phenomena simulations.

Experienced in **Enhanced Sampling methods:** Steered MD, MetaDynamics, Mean-Force Dynamics,... and Potential of Mean-Force (PMF) calculation.

Experienced in computational modeling of **crystalline** and **polymeric** molecular systems.

Experienced with **compiling** and **running** open-source packages (Lammps, Colvar, Plumed, OpenMPI,...) on high-performance computing (HPC) clusters. I was successful in deploying many packages on UNIST's clusters, note at this link.

Experienced with **developing pre- & post-processing codes**. One of my packages is at this link.

Knowledge of application Machine Learning in computational materials science.

Knowledge of Finite Element Method.

Programing languages: Python, Matlab, C++.

Miscelleneous: Lammps, Plumed, Ovito, Abaqus, Github, Pip, Conda, Jupyterlab, Matplotlib, Latex,...

PUBLICATIONS

Articles in my personal web.
Articles in my ORCID profile.
Articles in my Google Scholar profile.

Journal Articles

2022 Viet Hung Ho*, Cao Thang Nguyen*, Hoang D. Nguyen*, Hyun Suk Oh, Myoungsu Shin, Sung Youb Kim. "Hydrogenated Graphene with Tunable Poisson's Ratio Using Machine Learning: Implication for Wearable Devices and Strain Sensors." *ACS Applied Nano Materials*, 2022, 10617–10627. doi:10.1021/acsanm.2c01950. (* equal contribution)

2022 Viet Hung Ho*, Duc Tam Ho, Cao Thang Nguyen, Sung Youb Kim. "Negative out-of-plane Poisson's ratio of bilayer graphane." *Nanotechnology*, 33 255705. doi:10.1088/1361-6528/ac5da0.

2019 Cao Thang Nguyen*, Duc Tam Ho*, Seung Tae Choi, Doo-Man Chun, Sung Youb Kim. "Pattern transformation induced by elastic instability of metallic porous structures." *Computational Materials Science*, (2019) 17–24. doi:10.1016/j.commatsci.2018.10.023. (* equal contribution)

2019 Duc Tam Ho*, Cao Thang Nguyen*, Soon-Yong Kwon, Sung Youb Kim. "Auxeticity in Metals and Periodic Metallic Porous Structures Induced by Elastic Instabilities (Phys. Status Solidi B 1/2019)." *physica status solidi (b)*. doi:10.1002/pssb.201970010. (* equal contribution)

2019 Duc Tam Ho*, Cao Thang Nguyen*, Soon-Yong Kwon, Sung Youb Kim. "Auxeticity in Metals and Periodic Metallic Porous Structures Induced by Elastic Instabilities (Phys. Status Solidi B 1/2019)." *physica status solidi (b)*. doi:10.1002/pssb.201800122. (* equal contribution)

Manuscripts

2022 Cao Thang Nguyen, Sung Youb Kim. "Origami-inspired Graphene/PMMA composite with tunable auxetic property." *private arXiv*.

2022 Cao Thang Nguyen, Sung Youb Kim. "An enhanced sampling approach for computing the temperature-dependent free energy of solid surface and solid-liquid interface." *private arXiv*.

2021 Cao Thang Nguyen, Sung Youb Kim. "Mechanical vs. thermodynamic melting of metals from the mean-force dynamics calculation." *private arXiv*.

2021 Cao Thang Nguyen, Duc Tam Ho, Sung Youb Kim. "Coalescence-enhanced melting in the incipient stage of surface melting." *private arXiv*.

2020 Cao Thang Nguyen, Viet Hung Ho, Sung Youb Kim. "Finite-size effect on the thermodynamics melting predicted by free energy approach." *private arXiv*.

Codes/Tools

2021 Cao Thang Nguyen. "thatool: an object-oriented Python package for pre-processing and post-processing data from MD simulations.". *This package is avalibale for both PIP and Conda reposities, with the full documentation.*

Updated: October 2022